NewsBot Intelligence System

ITAI 2373 | Project Reflection | Zain Ahmed | Individual Work

1. Organization and collaboration context

This NewsBot implementation was completed as an individual project rather than a three- to four-person group collaboration. No team meetings, delegated coding tasks, or peer contributions are claimed. The work was organized into dataset validation, linguistic preprocessing, feature extraction, classification and model evaluation, syntax/entity analysis, and final results documentation. The notebook links these stages into a single execution path rather than isolated examples.

Working without a group made the experiment configuration easier to keep consistent, but it also meant there was no teammate to independently test results or review assumptions. Clear intermediate outputs, documented settings, and automated checks were therefore especially useful for reproducibility.

2. Technical integration and computational choices

The Colab notebook loaded 1,490 labeled BBC article rows and retained 1,440 usable unique articles. Cleaning and lemmatization prepared the classifier inputs, while a stratified split separated 1,080 training articles from 360 test articles. The TF-IDF vocabulary was fitted only on training examples, avoiding the data leakage that would result if test articles influenced the feature space.

The comparison included Multinomial Naive Bayes, Logistic Regression, and LinearSVC. Training-only cross-validation selected LinearSVC; its held-out accuracy was 96.39%, with macro F1 of 0.9608. An additional Bag-of-Words baseline offered a check on whether TF-IDF was actually improving classification. Because full syntactic parsing is more expensive than vectorizing text, POS, dependency, and named-entity analysis used 175 stratified articles while model training used the larger set.

3. How the NLP components work together

TF-IDF highlights words associated with categories, while classifiers use features to assign article labels. POS analysis measures grammatical patterns in the articles; dependency parsing extracts approximate who-did-what-to-whom relationships. Sentiment analysis provides a separate estimate of the positive or negative language used in an article. Named-entity recognition identifies people, organizations, locations, dates, and monetary expressions. The integrated inference function combines topic, sentiment, entity, and syntactic outputs for a single new input.

NewsBot Intelligence System | Reflection

Measured findings, business implications, and future work

4. Findings and business relevance

The confusion matrix showed that three technology articles were assigned to entertainment, the most common observed category mismatch. This is worth investigating because reporting about consumer technology, music services, and digital entertainment can combine the vocabularies of both categories. A newsroom could use the classifier for preliminary topic routing, while retaining human review for ambiguous content.

In the VADER analysis, politics had the largest negative-label share at 46.6%; sport and entertainment received positive labels for 78.9% and 80.2% of their articles. The POS sample showed that sport had the greatest proper-noun share, about 11.2%, consistent with coverage mentioning teams, athletes, and events. These are descriptive patterns in a historical news dataset, not evidence that readers or the public hold those feelings. Entity co-mentions likewise should not be mistaken for a verified relationship between people or organizations.

5. Individual participation and professional learning

I collected the labeled BBC training CSV and ran the integrated NewsBot notebook in Google Colab. I exported the model results and submission artifacts, then reviewed the accuracy figures and generated report. This gave me practical experience following an end-to-end NLP experiment, tracking a shared dataset through multiple analyses, and checking whether a model has actually completed its evaluation. AI tools assisted with code development and documentation; the exported results were generated by the Colab execution.

A key lesson is that a high classification score does not mean the system understands news perfectly. Errors can come from overlapping categories and unfamiliar current events. Likewise, extracting a name or calculating a sentiment score does not verify a story. For a useful business application, the system must make uncertainty, data provenance, and human review part of its workflow.

6. Future development

Future experiments could evaluate genuinely new and timestamped news coverage, measure error rates by individual category, compare domain-adapted sentence embeddings, and validate entity extraction against hand-annotated articles. A stronger sentiment evaluation would use labeled news examples rather than relying only on a lexical score. These extensions would show whether the system generalizes beyond the historical BBC sample and would better support production use.